

Advanced (Habanero)**Q1.** What is the expansion of $(a + b)^2$?

- (A) $a^2 + 2ab + b^2$
 (B) $a^2 - 2ab + b^2$
 (C) $a^2 + b^2$
 (D) $2a^2 + 2b^2$
 (E) $a^2 - b^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x - 4$
 (C) $x^2 + 2x + 1$
 (D) $x^2 - 2x - 1$
 (E) $x^2 + 1$

Q3. What is the factored form of $x^2 + 6x + 9$?

- (A) $(x + 3)(x - 3)$
 (B) $(x + 3)^2$
 (C) $(x - 3)^2$
 (D) $(x + 1)(x + 9)$
 (E) $(x - 1)(x - 9)$

Q4. What is the expansion of $(a - b)^2$?

- (A) $a^2 + 2ab + b^2$
 (B) $a^2 - 2ab + b^2$
 (C) $a^2 + b^2$
 (D) $2a^2 + 2b^2$
 (E) $a^2 - b^2$

Q5. Which of the following is not a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x - 4$
 (C) $x^2 + 2x + 1$
 (D) $x^2 + 2x + 2$
 (E) $x^2 + 1$

Q6. What is the factored form of $x^2 - 4x + 4$?

- (A) $(x + 2)(x - 2)$
 (B) $(x - 2)^2$
 (C) $(x + 2)^2$
 (D) $(x - 1)(x - 4)$
 (E) $(x + 1)(x + 4)$

Q7. What is the expansion of $(2x + 3)^2$?

- (A) $4x^2 + 12x + 9$
 (B) $4x^2 - 12x + 9$
 (C) $4x^2 + 9$
 (D) $12x^2 + 9$
 (E) $4x^2 + 12x - 9$

Q8. What is the factored form of $x^2 + 2x + 1$?

- (A) $(x + 1)(x - 1)$
 (B) $(x - 1)^2$
 (C) $(x + 1)^2$
 (D) $(x - 2)(x + 1)$
 (E) $(x + 2)(x - 1)$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 10x + 25$ and show all steps.**Q10.** Expand the expression $(3x - 2)^2$ and show all steps.**Chef's Tips**

- Tip 1: Make sure to check the signs of the terms in the perfect square trinomials.
- Tip 2: Use the formula $(a + b)^2 = a^2 + 2ab + b^2$ to expand binomials.
- Tip 3: Use the formula $(a - b)^2 = a^2 - 2ab + b^2$ to expand binomials.